



AgriSmart

.....

AgriSmart

Leveraging Modern Technology
to Transform Farm Management in Tunisia

MOBILE-FIRST · OFFLINE-CAPABLE · AI-POWERED · SCALABLE

► Presented By:
AgroSense



social media :
AgriSmart





Outline

1 Introduction & Context

2 Main Objectives

3 Target Users

4 Needs Analysis

5 Existing Solutions

6 Problem Statement

7 Proposed Solution

8 Innovation & added value

9 BMC

10 Alignment with SDGs

11 Global Architecture & final Outputs

12 Poster & Marketing video

13 DEMO Video

14 Conclusion & Perspectives





AgriSmart

Tunisia has the land. The farmers. The resources.

Yet production is declining, water is wasted,
crop losses are rising, and farmers are flying blind.



How can we leverage modern technology to its fullest
to improve agriculture and farm management in
Tunisia?





AgriSmart

Agriculture Powers Tunisia



~10% of GDP
comes from agriculture



500K+ jobs
depend on farming



**62% of national
water use**
goes to agriculture



**~80% smallholder
farmers**
drive production





AgriSmart

Why Tunisian Agriculture Is Under Pressure



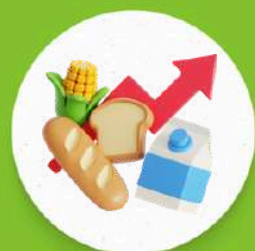
Low AgriTech investment



Severe water stress



Weak rural connectivity



Food prices are surging



Tunisia risks falling behind





What Does Success Look Like?



► Profitability



ooo

- Reduce operational waste (water, fertiliser, labour)
- Minimise crop loss through early detection
- Enable data-backed decisions at every stage
- Build a farm financial history for credit & investment access
- Optimise market timing with yield forecasting



IMPROVE RETURN ON INVESTMENT

► Productivity



ooo

- Increase yield per hectare through precision interventions
- Improve crop quality with real-time health monitoring
- Scale cattle output with automated health & feeding alerts
- Reduce time-to-decision for farm managers
- Enable consistent performance across seasons



IMPROVE QUALITY & QUANTITY OF OUTPUT



Who Are We Solving This For?

•▶ PRIMARY USER



Farm Owners & Managers

The person on the ground making daily decisions about crops, water, livestock, and labour. Ranges from smallholder farmers in rural Tunisia to managers of large commercial estates.

•▶ SECONDARY USERS



Agri Investors

Need portfolio visibility, ROI tracking and risk dashboards across multiple farms



Government Bodies

Seek national food security data, subsidy targeting and agricultural policy insights



AgriSmart

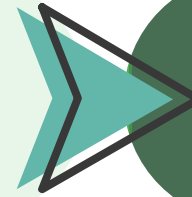


What Farm Managers Actually Need

SURFACE NEEDS — What they ask for

ROOT NEEDS — What they actually require

Better irrigation control



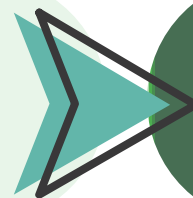
Stop losing money to water waste

Know when crops are sick



Prevent loss before it becomes visible

Track their cattle



Protect their most valuable assets automatically

See their farm data



Make confident decisions without guesswork

Reduce labour costs



Automate repetitive tasks so they focus on strategy



What Tunisia Has Tried So Far

Tunisian farmers haven't stood still — several agritech solutions have emerged. Each solves one piece. None solves the whole.

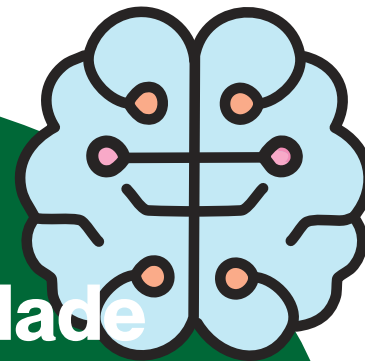
Solution	What it does	What it solves	What's still missing
Smart Farm	Soil moisture sensors + cloud platform	✓ Irrigation data	✗ No AI, no offline, no unified view
Seabex	Precision farming SaaS with satellite data	✓ Field monitoring	✗ Online only, no cattle, no automation
IrWise	Smart remote irrigation control via IoT	✓ Water control	✗ Single-purpose, no crop intelligence
Ezzarya	Sensors in irrigation pipes & soil	✓ Salinity & irrigation	✗ Hardware only, no mobile app
iFarming	Web & mobile apps using IoT and AI	✓ Partial monitoring	✗ No offline, fragmented features



The Gaps That Are Costing Tunisian Farms



Wasted Production



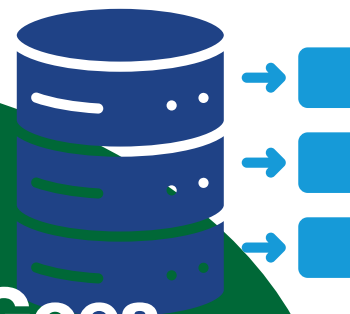
Decisions Made Blind



Timers Instead of Intelligence



Tools That Don't Reach Rural Areas



Data That Goes Nowhere



No Integrated Platform



THE CENTRAL PROBLEM

WHY HAS NO ONE EVER BROUGHT EXISTING TECHNOLOGY TOGETHER, DESIGNED FOR THE TUNISIAN FARMER, AND MADE IT WORK OFFLINE?

The technology to address every single one of these challenges already exists — it has simply never been unified into one solution built for this context.

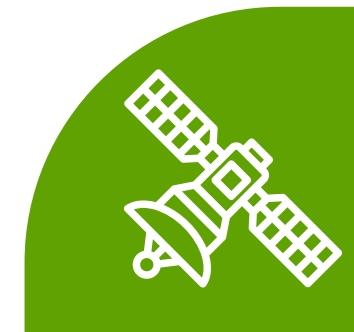
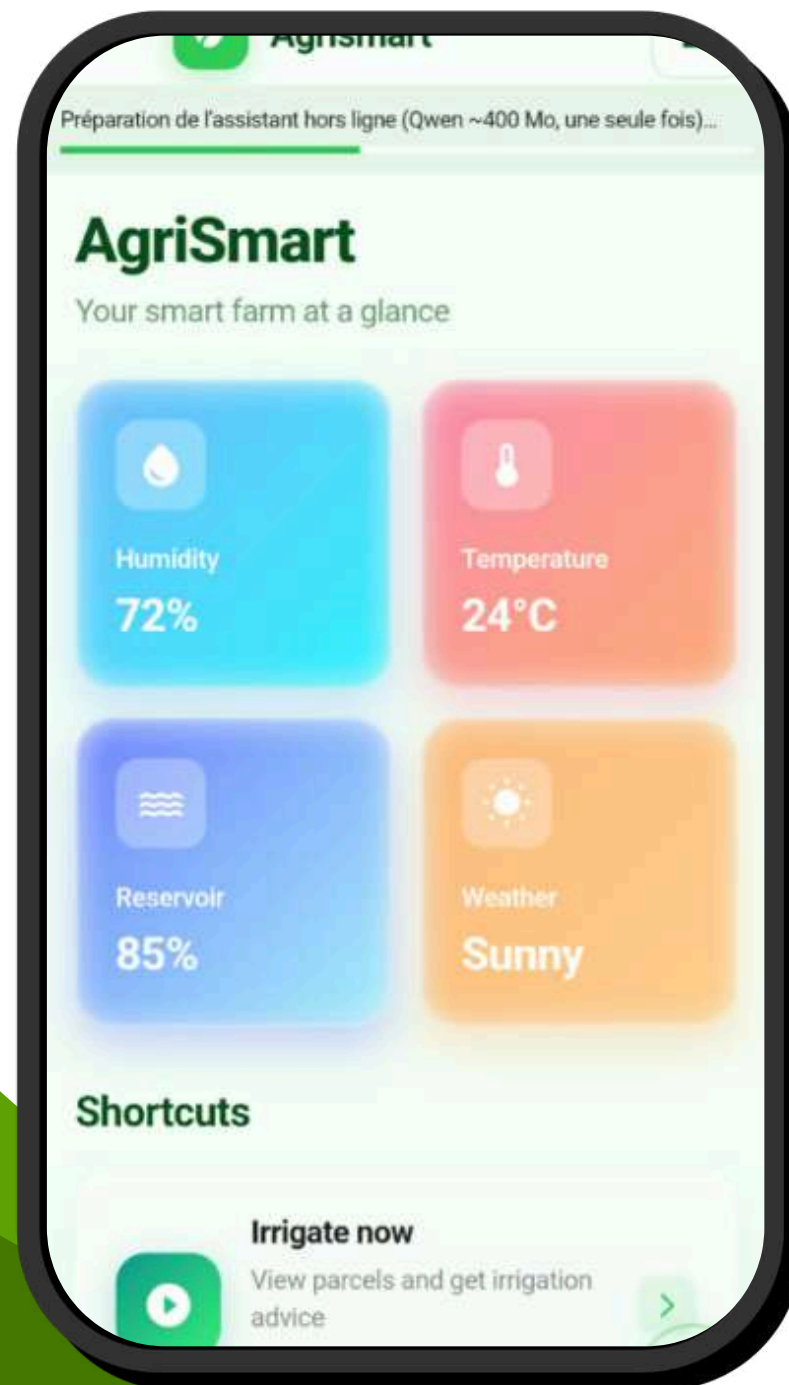
UNTIL NOW.



AgriSmart



AgriSmart — The All-in-One Farm Intelligence Platform



Satellite Field Analysis



AI Plant & Disease Detection



Autonomous Irrigation



Unified Dashboard



AgriSmart

From Features to Impact — The Three Value Pillars



ACCESSIBILITY

Offline app
simple UI
low-cost device support



AUTOMATION

AI agents
Sensors
Satellite triggers



PROFITABILITY

Less waste
Better timing
Bankable data

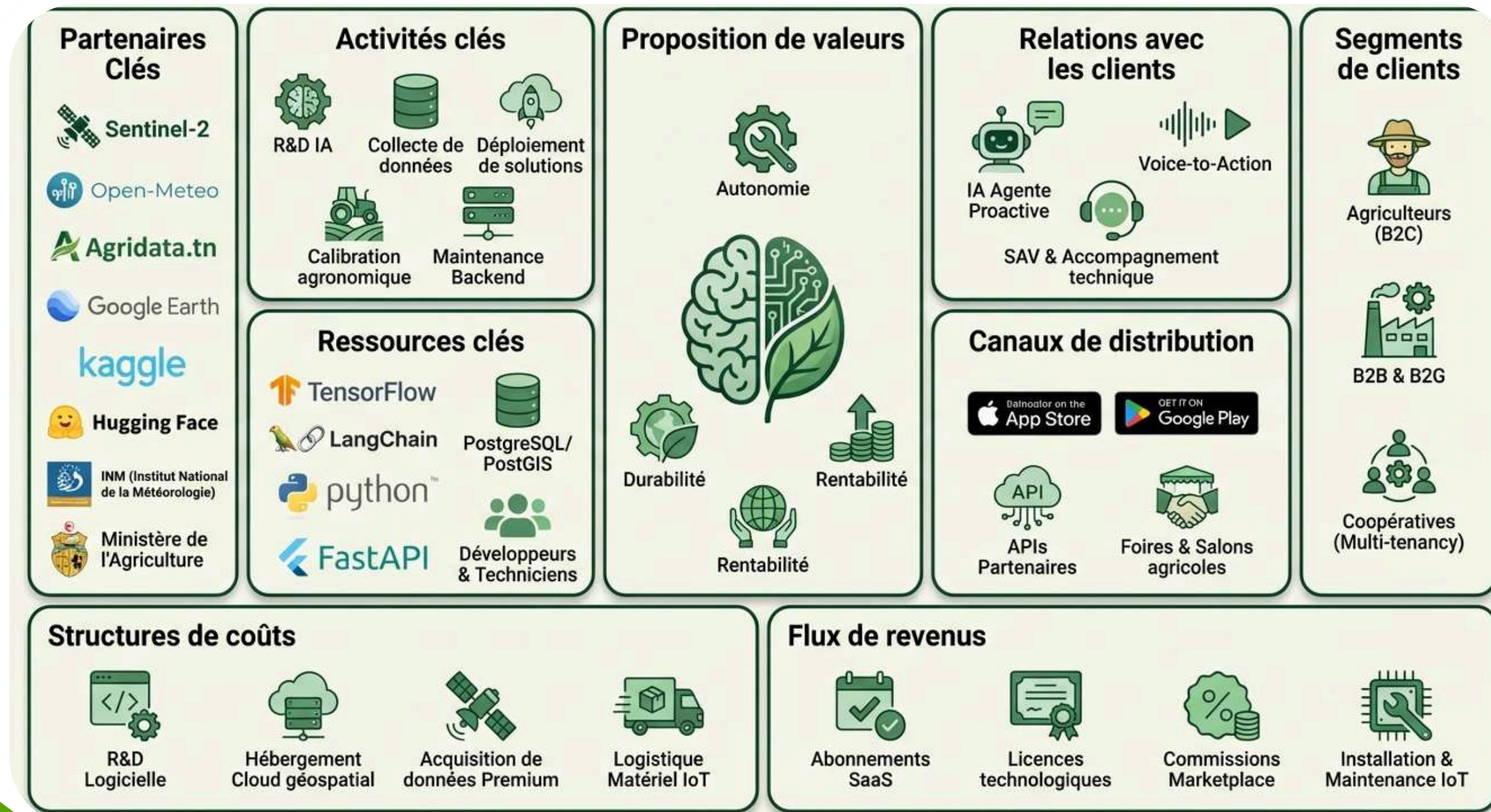


BUSINESS MODEL CANVAS

COMMENT?

QUOI ?

QUI?



COMBIEN ?



AgriSmart

SDG Alignment



Zero Hunger



Good health and
well being



Responsible Consumption
& production



Climate Action



TDSP Methodology



1

Business Understanding

2

Data Aquisition

3

Modeling

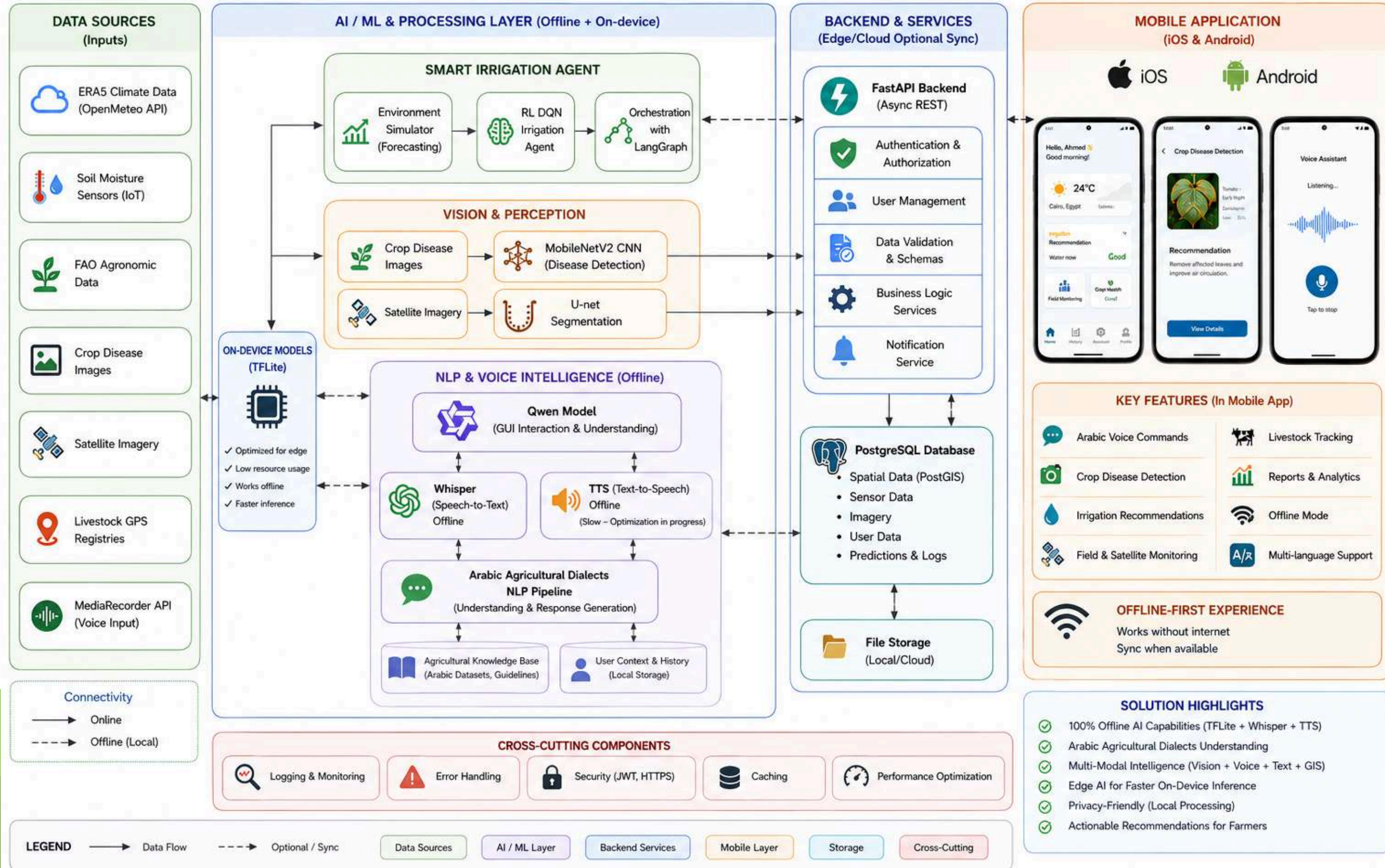
4

Deployment



AgriSmart

Global Architecture



MISSION: Empowering farmers with intelligent, accessible, and offline-first AI tools for sustainable agriculture.

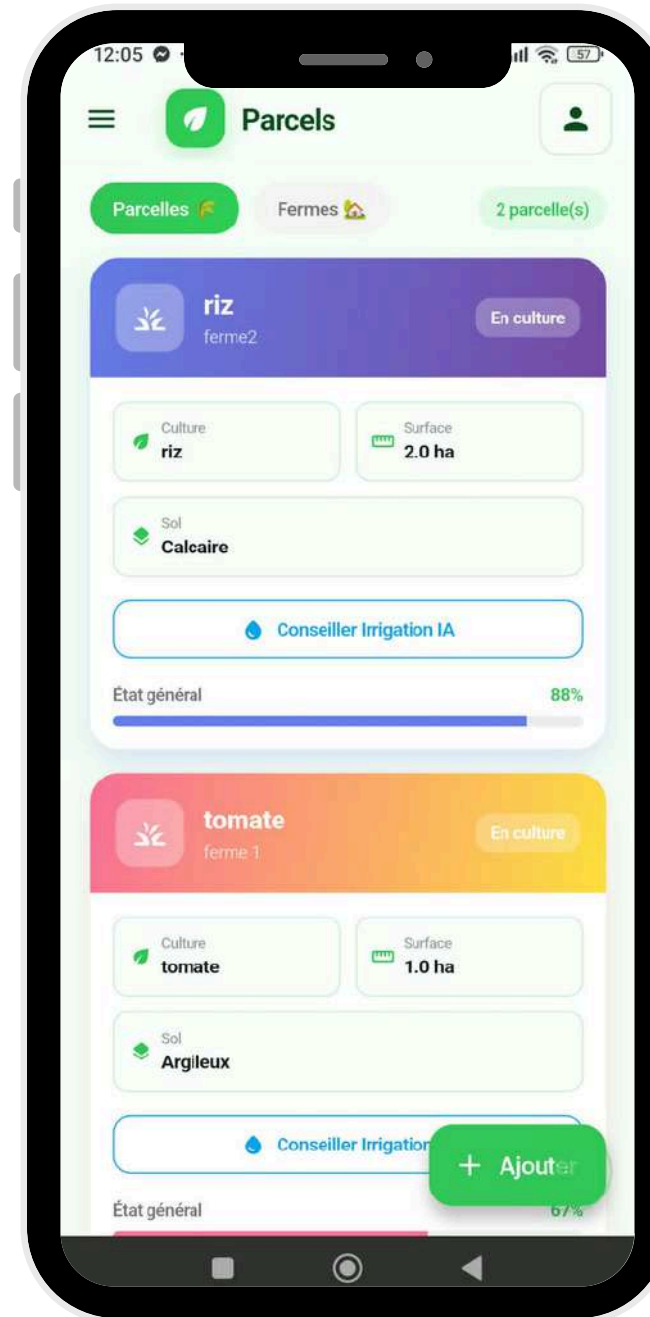
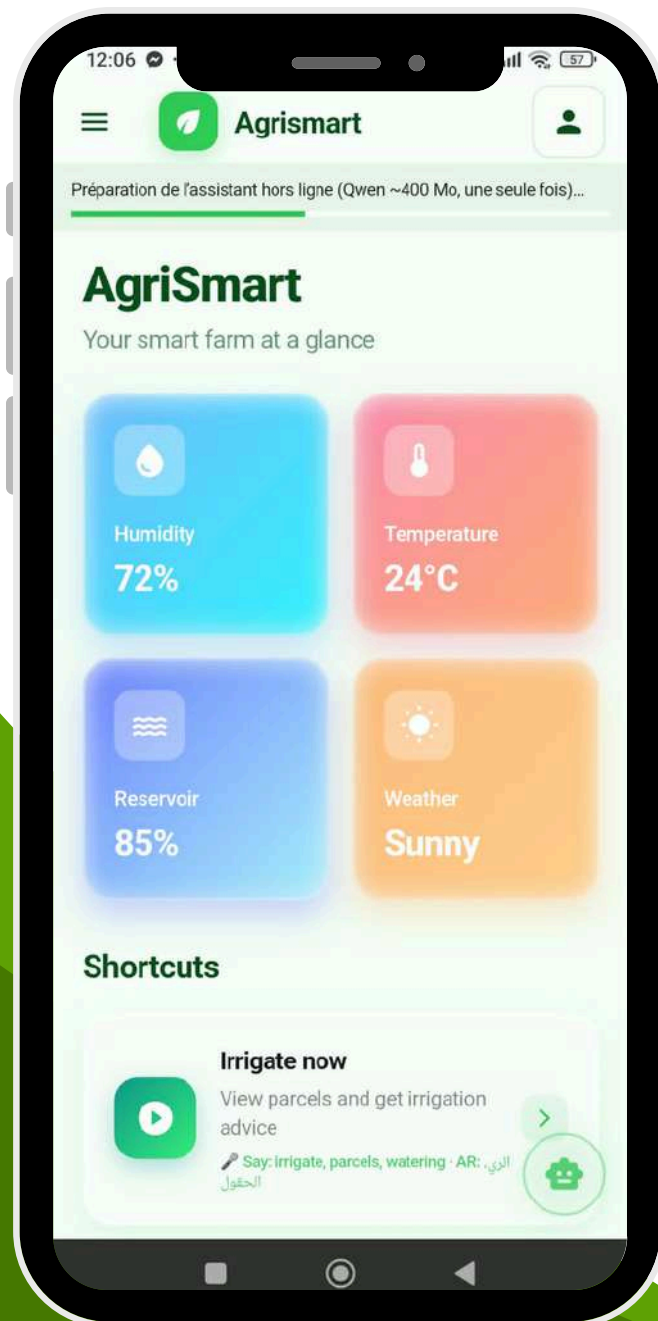


AgriSmart



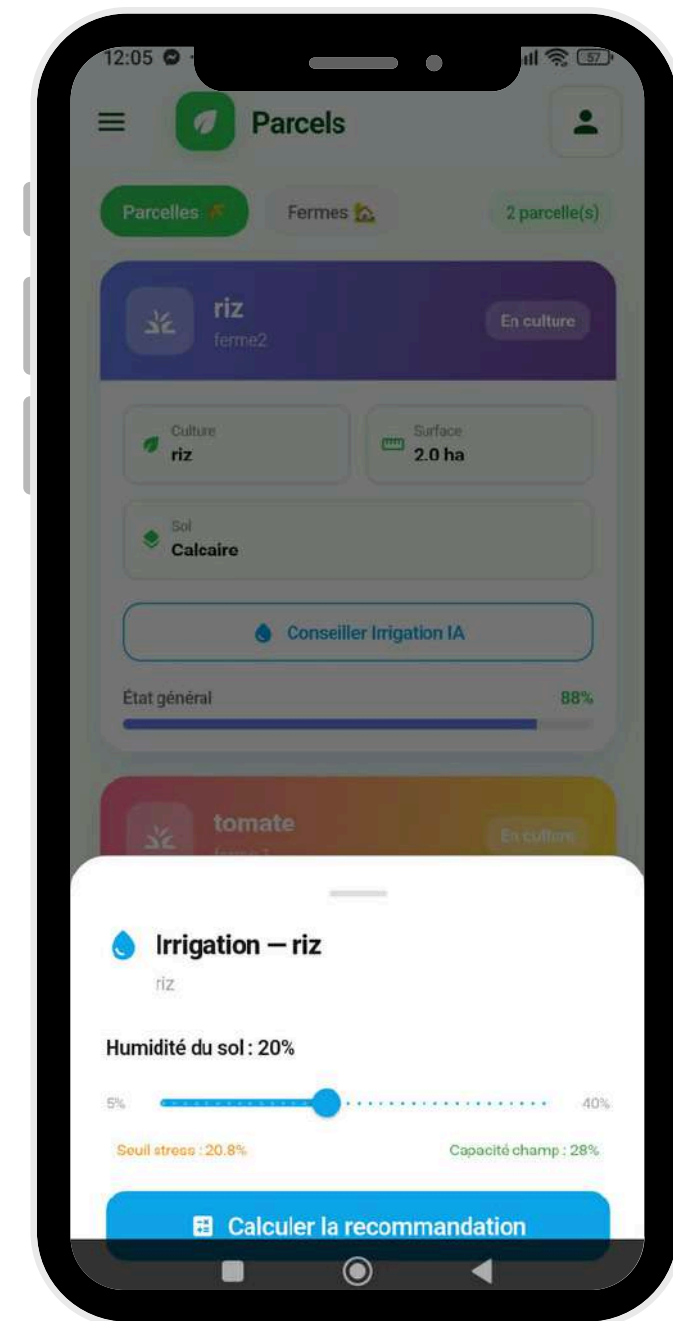
AgriSmart — Final Output

Live sensor dashboard —
humidity, temp, reservoir at a
glance



Parcel health scoring — AI
monitors each plot in real time

AI Irrigation Agent — soil
moisture → mm/day
recommendation

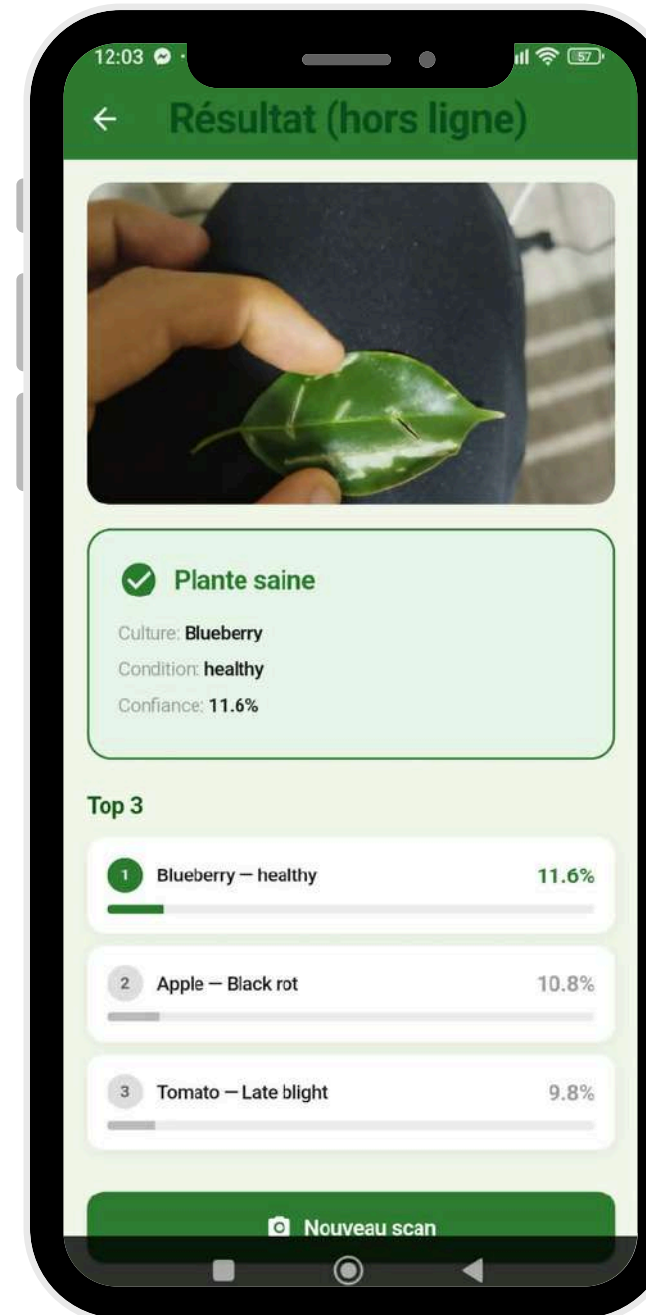
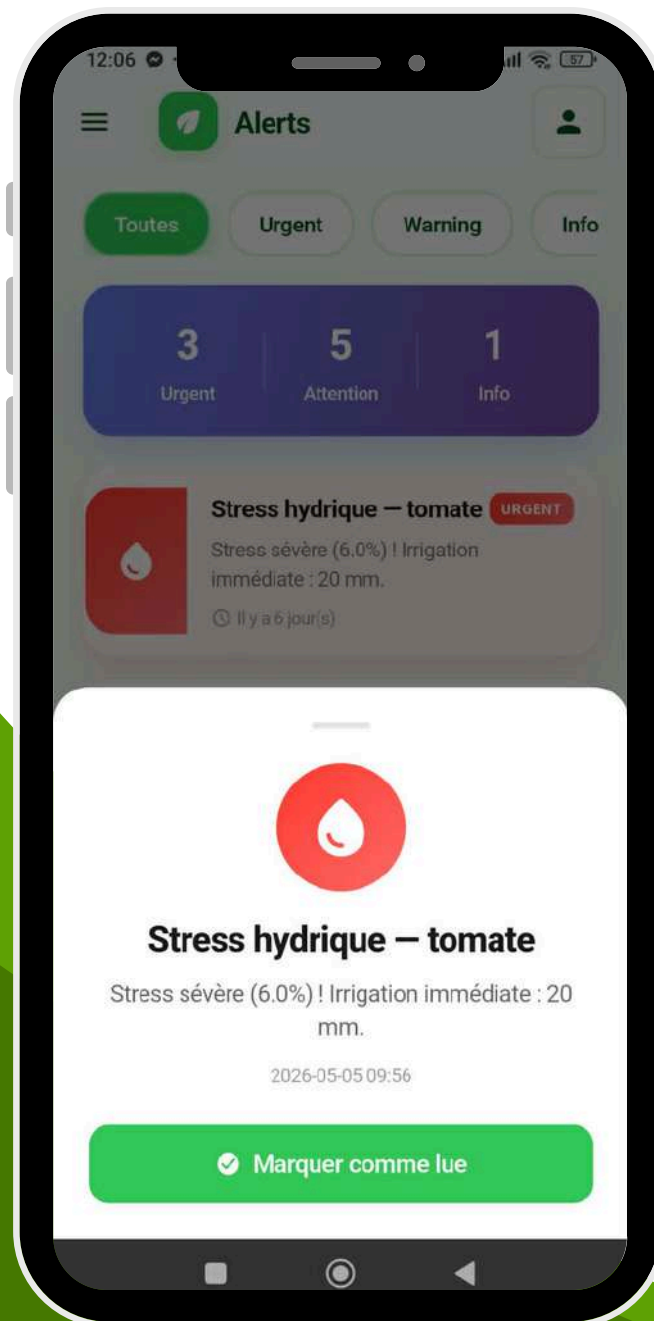




AgriSmart

AgriSmart — Final Output

Urgent alert — RL model detects stress, prescribes 20mm instantly



Offline disease scan — zero connectivity, result in <1s

AgriBot — auto-switches Groq (online) ↔ Qwen (offline)





AgriSmart



AGRISMART

Autonomous Agriculture Platform

Mobile-First · AI-Powered ·
Tunisia & MENA



Tunisian farmers lose up to 30% of their annual yield due to late disease detection, water mismanagement, and decisions still made on intuition — while existing tech is fragmented, cloud-only, and inaccessible in the field.

The Solutions

IoT+Satellite

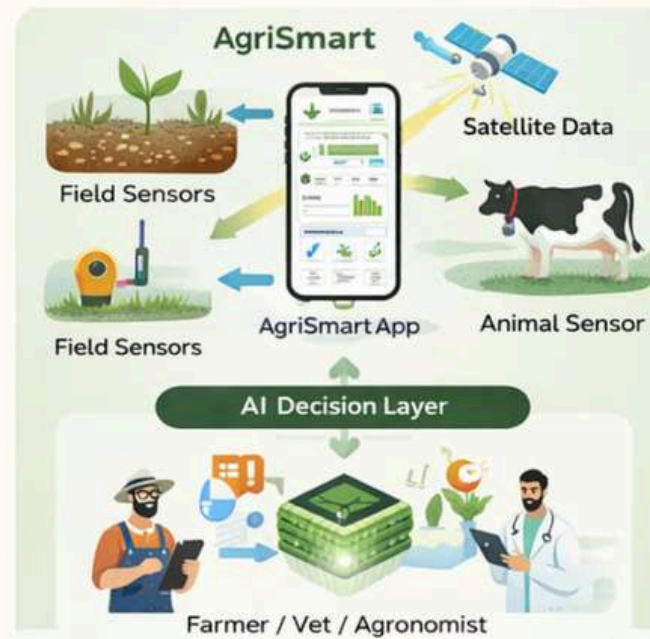
Soil sensors, weather data & NDVI imagery collected in real time

Agentic AI

AI agents reason, predict outcomes and trigger smart actions autonomously

Action on Mobile

Farmer receives clear alerts & recommendations — even offline



One unified app replaces fragmented tools: autonomous irrigation, offline crop disease scan, livestock health tracking and precision soil mapping — all in the farmer's pocket.

**GROW MORE, SPEND LESS —
WITHOUT GUESSING.**



Why AgriSmart?

"The only platform built offline-first for rural fields, with explainable AI agents — not black boxes — co-designed with local farmers, not for them."

Three Key Benefits

+25%

Crop yield increase through AI-optimized irrigation and early disease intervention

+40%

Water savings per season with autonomous sensor-driven irrigation agents

3 SEC

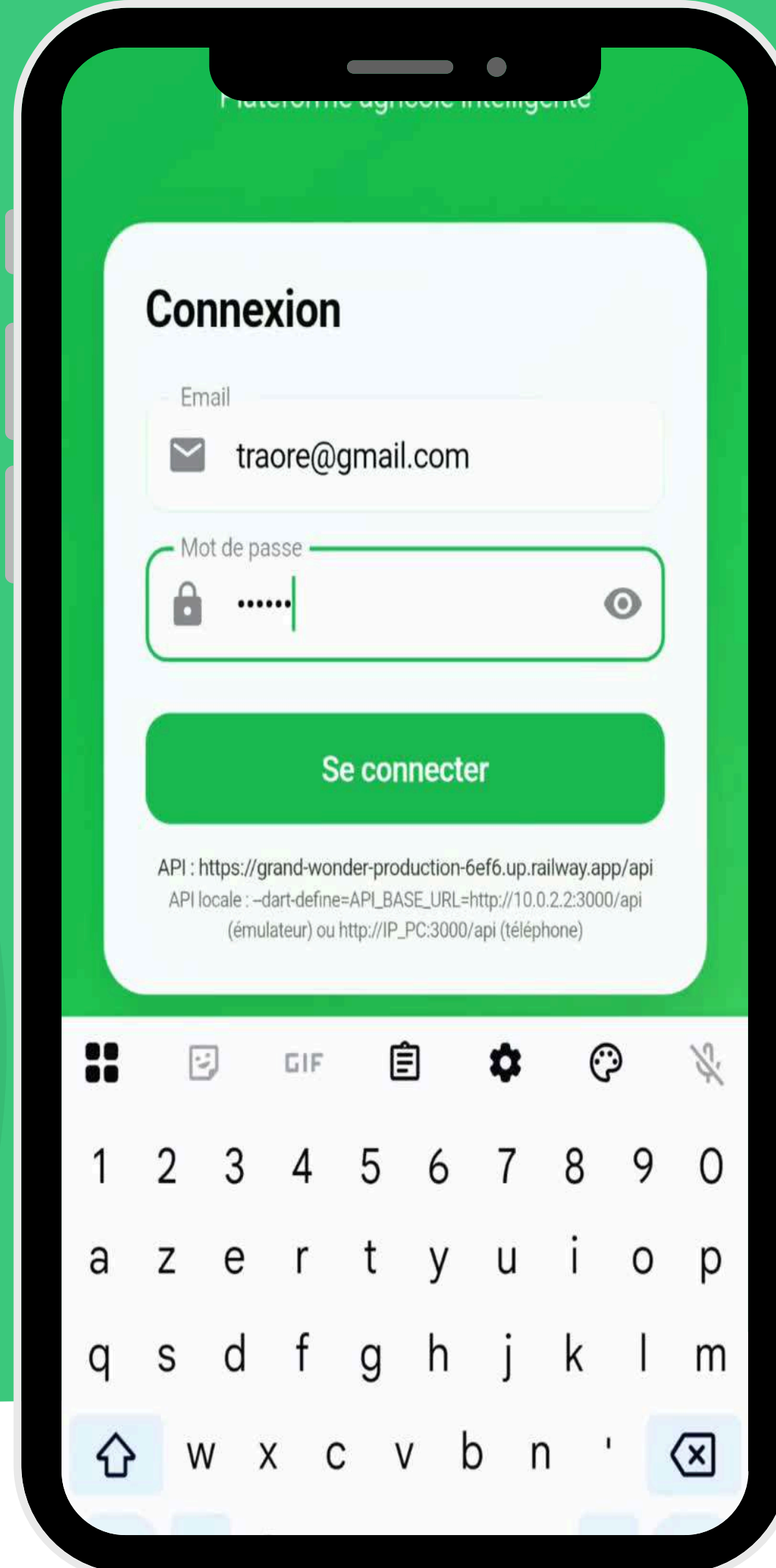
Disease diagnosis time — on-device, offline, directly from a photo on any smartphone

ONCE UPON

A TIME

Introduce Our New App

Agrismart





AgriSmart

Conclusion

Tunisia has the land, the farmers, and the resources — what was missing was a unified, intelligent, and accessible platform built for its reality.

AgriSmart answers that gap. By combining satellite field analysis, AI-powered disease detection, autonomous irrigation, and a mobile-first offline-capable interface, we deliver the first all-in-one farm intelligence platform designed specifically for the Tunisian agricultural context.

The result : less waste, better decisions, and bankable farm data — turning smallholder farmers from reactive to proactive, and from isolated to connected.



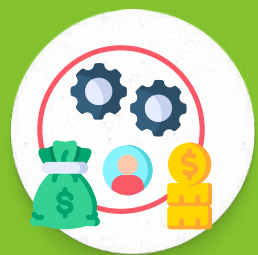


AgriSmart

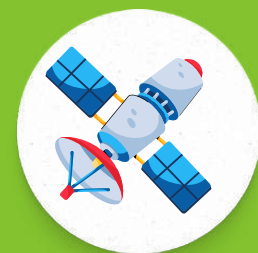
Future Perspectives



**Geographic
Expansion**



**Financial
Integration**



**Deeper Satellite
Intelligence**



**Government &
Policy Dashboard**

AgriSmart is not just a product — it is the infrastructure for a smarter, more resilient Tunisian agriculture. The technology is ready. The need is urgent. The moment is now.





AgriSmart

Thank You

For Your Attention

